

Acoustics of a Viscoelastic Liquid with Perfluorocarbon Droplets in the Presence of Phase Transitions

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Abstract—In this work, a modified Rayleigh–Lamb equation which takes into account the radial oscillations of a perfluorocarbon droplet with a vapor bubble in the center in a viscoelastic liquid has been derived. The viscoelasticity of the carrier medium has been taken into account based on the Zener rheological model. For small inclusion oscillations, the problem of heat and mass transfer between vapor, liquid phase and carrier liquid has been solved. The energy equation has been found. Based on the obtained equations of radial oscillations of the inclusion, energy and the well-known wave equation for a bubbly liquid, the dispersion relation has been determined. For a mixture of polydimethylsiloxane with droplets of octafluoropropane C_3F_8 and vapor bubbles in the center, the dependencies of the phase velocity and attenuation coefficient on the perturbation frequency have been plotted. The influence of the shear modulus and relaxation time of the elastic carrier liquid on the dynamics of acoustic waves has been illustrated.

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1. INTRODUCTION

In recent times, the study of emulsions with phase transitions in which vapor bubbles are formed inside liquid droplets under the impact of ultrasound is of great interest. This process is known as acoustic droplet vaporization [1]. The use of such emulsions has a vast potential in biomedicine [2, 3] since, as distinct from usual gas microbubbles, the liquid droplets can be fairly small and can quietly flow through thin capillaries. For example, when using sonographic visualization, the liquid droplets, that flow through the vascular system of tumors, can evaporate and be used as the contrast agents to obtain images of the internal tumor structure [4]. As the liquid droplets, perfluorocarbon droplets such as decafluorobutane C_4F_{10} , octafluoropropane C_3F_8 and dodecafluoropentane C_5F_{12} droplets are usually used [5, 6]. The choice of such matters is connected with the fact that these matters remain metastable at the physiologic temperature and cannot evaporate spontaneously. An acoustic energy is necessary to evaporate them. Usually, in order to prevent rapid droplet dissolution in the carrying liquid the droplets are covered by a polymeric shell [7].

In [8], the analytic solution to the problem of droplet vaporization was first obtained. In that study the Rayleigh–Plesset equation that describes the evolution of bubbles inside a superheated perfluorocarbon droplet located at the center of rigid pipe was derived. However, this model was obtained under a strong simplification, namely, the droplet vaporization rate was considered to be constant during the entire evaporation process. In [9, 10], heat transfer and the vaporization rate were described as functions of the temperature based on the approach used in simulation of a vapor bubble in usual liquid [11].

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The numerical calculations were compared with superfast optic measurements of the vapor-gas bubble growth. Important role of gas diffusion in the growth of inclusion was underlined. However, both models were obtained under the assumption that the vapor bubble is located in the infinite perfluorocarbon liquid medium. In [12] a mathematical model of dynamics of growth of a vapor bubble located inside a liquid droplet of finite dimension was proposed. It was found that the theoretical calculations of variation in the vapor bubble radius of C_5F_{12} , C_3F_8 , and C_4F_{10} droplets located in water and the experimental data are in good agreement. In [13, 14], the viscoelastic droplet shell was taken into account. It was shown that the presence of the elastic shell has the significant effect on the process of droplet vaporization. In [15–17], the influence of the rheological properties of carrying liquid on bubble dynamics was investigated. This paper examines the dynamics of acoustic waves in a viscoelastic medium with droplets of liquid containing vapor bubbles.

2. EQUATION OF RADIAL OSCILLATIONS

Let us consider a droplet of liquid with a vapor bubble at the center in an external viscoelastic liquid. The equations of continuity and momentum in a spherical coordinate system in the presence of central symmetry are written as follows

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_v \rho_v) = -\frac{\partial \rho_v}{\partial t}, \quad r < R_1, \quad (1)$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_k) = 0, \quad r > R_1, \quad k = e, l, \quad (2)$$

$$\rho_k \left(\frac{\partial u_k}{\partial t} + u_k \frac{\partial u_k}{\partial r} \right) = -\frac{\partial p_k}{\partial r}, \quad r > R_1, \quad k = e, l. \quad (3)$$

Here ρ is the density, r is the radial coordinate, t is time, u is radial velocity, p is pressure, R_1 is the radius of the vapor bubble, R_2 is the radius of the droplet, the index v refers to the vapor, e refers to the droplet, l refers to the carrier liquid. The solutions to equations (1) and (2) have the form

$$u_v = \frac{r}{R_1} U_v, \quad u_k = \frac{R_1^2}{r^2} U_k, \quad k = e, l, \quad (4)$$

where U_v and U_k are, respectively, the velocities of vapor and liquid at the boundaries of the inclusion. Boundary conditions are written at the boundary R_1 taking into account phase transitions and R_2 without phase transitions

$$p_e(R_1) - p_v(R_1) + \frac{2\sigma_1}{R_1} = \tau_e(R_1) - \tau_v(R_1) + J(u_e(R_1) - u_v(R_1)),$$

$$J = \rho_v(\dot{R}_1 - U_v) = \rho_e(\dot{R}_1 - U_e), \quad (5)$$

$$p_l(R_2) - p_e(R_2) + \frac{2\sigma_2}{R_2} = \tau_l(R_2) - \tau_e(R_2), \quad u_e(R_2) = u_l(R_2) = \dot{R}_2. \quad (6)$$

Here σ is the surface tension coefficient, τ are the normal stresses, J is the intensity of phase transitions. Normal stresses for vapor and liquid droplet are given in the form [18, 19]

$$\tau_v = 2\mu_v \frac{\partial u_v}{\partial r} - \frac{2\mu_v}{3r^2} \frac{\partial(r^2 u_v)}{\partial r}, \quad \tau_e = 2\mu_e \frac{\partial u_e}{\partial r},$$

where μ is dynamic viscosity. Since the viscosity of the vapor is less than the viscosity of the liquid, therefore, $\tau_v \ll \tau_e$ and consequently, the value τ_v can be neglected in the future. The viscoelasticity of the carrier medium will be taken into account using the Zener rheological model

$$\lambda_l \dot{\tau}_l + \tau_l = 2G_l \varepsilon_l + 2\mu_l \dot{\varepsilon}_l, \quad (7)$$

where G is the shear modulus, λ is the relaxation time of the elastic phase. Taking into account (4), expression can be written in the form

$$\lambda_l \dot{\tau}_l + \tau_l = -\frac{4\mu_l}{r^3} \left(R_2^2 \dot{R}_2 + \frac{G_l}{3\mu_l} (R_2^3 - R_{20}^3) \right).$$

Let us write the solution of this equation as $\tau_l = -\frac{4\mu_l}{r^3} D_l$, then

$$\lambda_l \dot{D}_l + D_l = R_2^2 \dot{R}_2 + \frac{G_l}{3\mu_l} (R_2^3 - R_{20}^3). \quad (8)$$

The momentum equation (3) is integrated from R_1 to infinity. In this case, the interval from R_1 to R_2 will correspond to the parameters of the liquid droplet, from R_2 to infinity—to the parameters of the carrier liquid. Then, taking into account (4)–(8), we can obtain the equation of radial oscillations of the inclusion under consideration

$$\begin{aligned} \rho_d R_1 \dot{U}_e - 0.5 \rho_{d2} U_e^2 + 2 \rho_d \dot{R}_1 U_e = p_v - p_\infty - \frac{2\sigma_1}{R_1} - \frac{2\sigma_2}{R_1} + \left(\frac{1}{\rho_v} - \frac{1}{\rho_e} \right) J^2 \\ - \frac{4\mu_e}{R_1} \left(1 - \frac{R_1^3}{R_2^3} \right) U_e - \frac{4\mu_l}{R_2^3} D_l, \quad U_e = \dot{R}_1 - \frac{J}{\rho_e}, \\ \rho_d = \rho_e + (\rho_l - \rho_e) \frac{R_1}{R_2}, \quad \rho_{d2} = \rho_e + (\rho_l - \rho_e) \frac{R_1^4}{R_2^4}. \end{aligned} \quad (9)$$

Let us linearize equation (8) with respect to the equilibrium state: $R_2 = R_{20} + R'_2$ and taking into account the incompressibility condition

$$R'_2 = \frac{R_{10}^2}{R_{20}^2} \left(1 - \frac{J_0}{i\omega\rho_e} \right) R'_1,$$

then

$$\lambda_l \dot{D}_l + D_l = R_{10}^2 \left(1 - \frac{J_0}{i\omega\rho_e} \right) \dot{R}'_1 + R_{10}^2 \frac{G_l}{\mu_l} \left(1 - \frac{J_0}{i\omega\rho_e} \right) R'_1.$$

Let us present the solution to this equation in the form: $D_l = aR'_1 + b\dot{R}'_1$, then we can write

$$\lambda_l b \ddot{R}'_1 + \left(\lambda_l a + b - \left(1 - \frac{J_0}{i\omega\rho_e} \right) R_{10}^2 \right) \dot{R}'_1 + \left(a - \left(1 - \frac{J_0}{i\omega\rho_e} \right) \frac{R_{10}^2 G_l}{\mu_l} \right) R'_1 = 0.$$

We will look for a solution of this equation in the form of harmonic waves $R'_1 = \Delta R \exp(i\omega t)$, in this case we obtain

$$-b\lambda_l \omega^2 + a - \frac{R_{10}^2 G_l}{\mu_l} + \frac{J_0 R_{10}^2}{\rho_e} + i\omega \left(\lambda_l a + b - R_{10}^2 - \frac{1}{\omega^2} \frac{R_{10}^2 G_l J_0}{\mu_l \rho_e} \right) = 0.$$

To determine the unknown coefficients a and b , it is necessary to solve a system of equations

$$\begin{cases} -b\lambda_l \omega^2 + a - \frac{R_{10}^2 G_l}{\mu_l} + \frac{J_0 R_{10}^2}{\rho_e} = 0, \\ a\lambda_l + b - R_{10}^2 - \frac{1}{\omega^2} \frac{R_{10}^2 G_l J_0}{\mu_l \rho_e} = 0. \end{cases}$$

After solving this system of equations, we find

$$\begin{aligned} a &= \frac{R_{10}^2}{1 + \lambda_l^2 \omega^2} \left(\lambda_l \omega^2 + \frac{G_l}{\mu_l} - \left(1 - \frac{\lambda_l G_l}{\mu_l} \right) \frac{J_0}{\rho_e} \right), \\ b &= \frac{R_{10}^2}{1 + \lambda_l^2 \omega^2} \left(1 - \frac{\lambda_l G_l}{\mu_l} + \frac{\lambda_l J_0}{\rho_e} + \frac{1}{\omega^2} \frac{G_l J_0}{\mu_l \rho_e} \right). \end{aligned}$$

In a similar way, we linearize the equation (9): $R_1 = \Delta R \exp(i\omega t)$, $J = J_0 \Delta R \exp(i\omega t)$, we get

$$\begin{aligned} -\omega^2 \Delta R - \frac{i\omega J_0}{\rho_e} \Delta R + \frac{4\mu_e}{\rho_{d0} R_{10}^2} \left(1 - \frac{R_{10}^3}{R_{20}^3} \right) i\omega \Delta R - \frac{4\mu_e}{\rho_{d0} R_{10}^2} \left(1 - \frac{R_{10}^3}{R_{20}^3} \right) \frac{J_0}{\rho_e} \Delta R = \frac{\Delta p_v - \Delta p_\infty}{\rho_{d0} R_{10}} \\ - \frac{1}{\rho_{d0} R_{10}^2} \left(-\frac{2\sigma_1}{R_{10}} - \frac{2\sigma_2 R_{10}^3}{R_{20}^4} \right) \Delta R - \frac{2\sigma_2 R_{10}^3 J_0}{i\omega \rho_e \rho_{d0} R_{10}^2 R_{20}^4} \Delta R - \frac{4\mu_l (a + i\omega b)}{\rho_{d0} R_{10} R_{20}^3} \Delta R. \end{aligned} \quad (10)$$

Here the index 0 refers to the parameters of the equilibrium state. In the case when there are no phase transitions $J = J' = 0$, equations (9) and (10) take the form [20].

3. INTERFACIAL HEAT AND MASS TRANSFER

Thermal conductivity equations are written for vapor and liquid phase [18, 21] to take into account interfacial heat and mass transfer

$$\rho_v c_{pv} \frac{\partial T_v}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\lambda_v r^2 \frac{\partial T_v}{\partial r} \right) + \frac{\partial p_v}{\partial t}, \quad 0 < r < R_1, \quad (11)$$

$$\rho_e c_{pe} \frac{\partial T_e}{\partial t} = \lambda_e \frac{\partial^2 T_e}{\partial r^2}, \quad R_1 < r < R_2, \quad (12)$$

where T is temperature, c_p is specific heat, λ is thermal conductivity coefficient. Here and below, it is assumed that the thickness of the internal layer of the liquid phase of a droplet is much smaller than the size of the vapor bubble, therefore, the thermal conductivity equation (12) is written in a plane one-dimensional approximation. In the absence of external heat supply, the average mass temperature of the carrier liquid will be considered constant. In equation (11), pressure p_v is considered only as a function of time $p_v(t)$, thereby the homobaricity condition is satisfied [18]. This assumption will allow us to avoid solving the equation of vapour motion inside the bubble and thereby simplify the system of equations. When setting up the problem, it is necessary to set boundary conditions, which for temperatures have the form

$$r = 0: \quad \frac{\partial T_v}{\partial r} = 0, \quad r = R_1: \quad T_v = T_e = T_a, \quad J = \frac{1}{L} \left(\lambda_e \frac{\partial T_e}{\partial r} - \lambda_v \frac{\partial T_v}{\partial r} \right), \quad (13)$$

$$r = R_2: \quad T_l = T_0 = \text{const.} \quad (14)$$

It should be noted that a quasi-equilibrium phase transition scheme is adopted here, as evidenced by the condition that the temperature on the surface of the vapor bubble is equal to the saturation temperature T_a . Thus, the intensity of phase transitions J is determined by the last relation (13) from the solution of the thermal conductivity equation for liquid (12) and vapor (11). The saturation temperature T_a is determined from the Clapeyron–Clausius equation

$$T_a = \left(\frac{1}{T_{a0}} - \frac{B_v}{L} \ln \left(\frac{p_v}{p_\infty} \right) \right)^{-1}. \quad (15)$$

Here T_{a0} is the temperature at the saturation line, which depends on the type of liquid, B_v is the reduced vapor constant, L is the specific heat of vaporization of the liquid. The solution to the above problem (11)–(15) for the case of small oscillations of the inclusion with small amplitude is determined in the form of the real parts of complex functions

$$T_j = T_0 + T'_j \exp(i\omega t), \quad j = v, e, \quad p_v = p_\infty + p'_v \exp(i\omega t), \quad (16)$$

where ω is the frequency of disturbances. Equation (11), taking into account a solution of the form (16), will be written as follows

$$\frac{\partial^2 T'_v}{\partial r^2} + \frac{2}{r} \frac{\partial T'_v}{\partial r} - \frac{i\omega}{\kappa_v} T'_v = -\frac{i\omega}{\lambda_v} p'_v, \quad \kappa_v = \frac{\lambda_v}{\rho_v c_{pv}}. \quad (17)$$

The general solution of the second-order differential equation (17) is found as the sum of a particular solution $T'_v = \kappa_v p'_v / \lambda_v$ and a linear combination of two independent solutions of the homogeneous equation

$$T'_v = \frac{\kappa_v}{\lambda_v} \left(1 + V_1 \frac{\sinh(rY_v)}{r} + V_2 \frac{\cosh(rY_v)}{r} \right), \quad Y_v = \sqrt{\frac{i\omega}{\kappa_v}},$$

where V_1 and V_2 are unknown variables. The solution to equation (12) can be written in a similar way

$$T'_e = E_1 \sinh[(r - R_{10})Y_e] + E_2 \cosh[(r - R_{10})Y_e], \quad Y_e = \sqrt{i\omega/\kappa_e}.$$

The unknown variables V_1 , V_2 , E_1 , and E_2 are determined from the boundary conditions (13)–(15). Thus, analytical expressions are determined for the temperature distribution of vapor T_v , droplet of liquid

T_e and intensity of phase transitions J . At the next stage, the energy equation for the vapor bubble is derived. The Clapeyron–Mendeleev equation is used as the equation of state of vapor

$$p_v = \rho_v B_v T_v = \frac{\gamma - 1}{\gamma} c_{pv} \rho_v T_v, \quad (18)$$

where γ is the adiabatic exponent. Equation (18) is differentiated with respect to time t and, taking into account (1), takes the form

$$\frac{dp_v}{dt} = \frac{\gamma - 1}{\gamma} \rho_v c_{pv} \frac{dT_v}{dt} - \frac{2}{r} p_v u_v. \quad (19)$$

The heat inflow equation for the vapor phase (11) is integrated over the radial coordinate from zero to R_{10} , linearized and, taking into account (19), written as follows

$$\frac{dp'_v}{dt} = -\frac{3\gamma p_{v0}}{R_{10}} U'_v + 3(\gamma - 1) \frac{\lambda_v}{R_{10}} \left(\frac{\partial T'_v}{\partial r} \right)_{r=R_{10}}, \quad U'_v = \dot{R}'_1 - \frac{J'}{\rho_v}.$$

Taking into account the found expressions for T'_v and J' this equation will finally take the form

$$\frac{dp'_v}{dt} = -\frac{3\gamma p_{v0}}{R_{10}} \frac{dR'_1}{dt} - i\omega \Psi p'_v, \quad (20)$$

$$\Psi = 3(\gamma - 1) \frac{y_v \coth y_v - 1}{y_v^2} \left(1 - \frac{\lambda_v \tau_v}{\kappa_v} \right) \left(1 - \frac{\gamma}{\gamma - 1} \frac{p_{v0}}{\rho_v L} \right) - 3i \frac{\gamma p_{v0}}{\rho_v R_{10}} \frac{\lambda_e y_e \coth y_e}{L r_e} \frac{\tau_v}{\omega},$$

$$\tau_v = \frac{B_v}{p_{v0} L} \left(\frac{1}{T_{a0}} - \frac{B_v}{L} \ln \left(\frac{p_{v0}}{p_\infty} \right) \right)^{-2}, \quad y_v = \sqrt{\frac{i\omega R_{10}^2}{\kappa_v}}, \quad y_e = \sqrt{\frac{i\omega r_e^2}{\kappa_e}}, \quad r_e = R_{20} - R_{10}.$$

Here r_e is the thickness of the internal layer of the liquid phase.

4. DISPERSION EQUATION

Let us write the solution to equation (20) in the form of progressive waves $R'_1 = \Delta R_1 \exp(i\omega t)$, $p'_v = \Delta p_v \exp(i\omega t)$, where ΔR_1 and Δp_v are the amplitudes of the waves. Then, taking into account (10), we determine from the obtained system of equations ΔR_1

$$\Delta R_1 = -\frac{\Delta p_\infty}{\rho_{d0} R_{10} (\omega_0^2 + i\omega\delta - \omega^2)}, \quad (21)$$

$$\omega_0^2 = X + \frac{p_{v0} \Re \Phi}{\rho_{d0} R_{10}^2} + \frac{\omega \Im J_0}{\rho_e} - \frac{4\mu_e}{\rho_{d0} R_{10}^2} \left(1 - \frac{R_{10}^3}{R_{20}^3} \right) \frac{\Re J_0}{\rho_e} - \frac{4\mu_l \xi_2 \Re J_0}{\rho_{d0} R_{10} R_{20}^3} - \frac{B \Im J_0}{\omega \rho_e},$$

$$\delta = -\frac{\Re J_0}{\rho_e} \left(1 - \frac{B}{\omega^2} \right) + \frac{p_{v0} \Im \Phi}{\rho_{d0} R_{10}^2 \omega} + \frac{4\mu_e}{\rho_{d0} R_{10}^2} \left(1 - \frac{R_{10}^3}{R_{20}^3} \right) \left(1 - \frac{\Im J_0}{\omega \rho_e} \right) + \frac{4\mu_l}{\rho_{d0} R_{10} R_{20}^3} \left(\xi_3 - \frac{\xi_2 \Im J_0}{\omega} \right),$$

$$J_0 = \frac{p_{v0} \Phi}{L R_{10}} \left(\frac{\lambda_e y_e}{r_e} \coth y_e \tau_v - i\omega R_{10} \frac{y_v \coth y_v - 1}{y_v^2} \left(1 - \frac{\lambda_v \tau_v}{\kappa_v} \right) \right), \quad \Phi = \frac{3\gamma}{1 + \Psi},$$

$$X = \frac{1}{\rho_{d0} R_{10}^2} \left(-\frac{2\sigma_1}{R_{10}} - \frac{2\sigma_2 R_{10}^3}{R_{20}^4} + \frac{4\mu_l R_{10} \xi_1}{R_{20}^3} \right), \quad B = \frac{1}{\rho_{d0} R_{10}^2} \left(-\frac{2\sigma_2 R_{10}^3}{R_{20}^4} + \frac{4\mu_l R_{10}}{R_{20}^3} \rho_e \xi_4 \omega^2 \right),$$

$$p_{v0} = p_\infty + \frac{2\sigma_1}{R_{10}} + \frac{2\sigma_2}{R_{20}}, \quad \xi_1 = \frac{R_{10}^2}{1 + \lambda_l^2 \omega^2} \left(\lambda_l \omega^2 + \frac{G_l}{\mu_l} \right), \quad \xi_2 = \frac{R_{10}^2}{\rho_e (1 + \lambda_l^2 \omega^2)} \left(1 - \frac{\lambda_l G_l}{\mu_l} \right),$$

$$\xi_3 = \frac{R_{10}^2}{1 + \lambda_l^2 \omega^2} \left(1 - \frac{\lambda_l G_l}{\mu_l} \right), \quad \xi_4 = \frac{R_{10}^2}{1 + \lambda_l^2 \omega^2} \left(\frac{\lambda_l}{\rho_e} + \frac{G_l}{\mu_l \rho_e \omega^2} \right).$$

In the case when there is only a vapor bubble in the liquid, the expressions for δ and ω_0 take the form [22]. Let us consider the propagation of an acoustic wave in a viscoelastic medium when it contains many droplets of liquid chaotically located along the wavelength, inside of which there are vapor bubbles. We will assume that the distance between inclusions is much greater than their size. As noted above, the size of the internal liquid layer satisfies the condition $r_e \ll R_{10}$. In this case, account of multiple wave scattering can be done in the usual way for a bubble medium based on the well-known Foldy-type wave equation in the long-wave approximation [23]

$$\frac{1}{C_l^2} \frac{\partial^2 p_\infty}{\partial t^2} - \frac{\partial^2 p_\infty}{\partial x^2} = 4\pi\rho_l \int_0^\infty R_{10}^2 \ddot{R}_1 f dR_{10}. \quad (22)$$

Here f is the function of distribution of inclusions by size and C_l is the speed of sound in the carrier liquid. Taking into account the fact that $\ddot{R}_1 = -\omega^2 \Delta R_1$, $\partial^2 p_\infty / \partial t^2 = -\omega^2 \Delta p_\infty$, and expression (21) the wave equation (22) will be rewritten in the form

$$\frac{\partial^2 \Delta p_\infty}{\partial x^2} + K_*^2 \Delta p_\infty = 0, \quad \frac{K_*^2}{\omega^2} = \frac{1}{C_l^2} + 4\pi\rho_l \int_0^\infty \frac{R_{10} f}{\rho_{d0} (\omega_0^2 + i\omega\delta - \omega^2)} dR_{10}, \quad (23)$$

where K_* is the complex wave number and the second equation (23) is the dispersion equation. If we choose $f = n\delta(R - R_{10})$, where δ is the Dirac function, $n = 3\alpha/4\pi R_{10}^3$ is the number of inclusions per unit volume of liquid, then the integral in (23) can be omitted and the inclusions can be considered monodispersed.

5. CALCULATION RESULTS

Let us consider the propagation of an acoustic wave in a viscoelastic liquid with droplets of octafluoropropane and vapor bubbles in the center at temperature $T_0 = 310$ K and pressure $p_0 = 0.1$ MPa. Polydimethylsiloxane (PDMS) was chosen as the carrier medium. Parameters of polydimethylsiloxane: $C_l = 1020$ m/s, $\rho_l = 965$ kg/m³, $\mu_l = 0.4 \times 10^6$ Pa s, $\lambda_l = 0.4$ s. The volume content of inclusions is $\alpha = 0.001$, the radius of vapor bubbles is $R_{10} = 7 \times 10^{-6}$ m, the thickness of droplets of liquid is $r_e = 10^{-6}$ m. Calculations were carried out using the dispersion equation (23). Thermophysical parameters of substances DDFP and OFP were taken from works [24, 25].

Figure 1 shows the results of a comparison of the dependencies of the phase velocity $C_p = \omega/\Re K_*$ (a) and the attenuation coefficient $K_{**} = \Im K_*$ (b) for a mixture of polydimethylsiloxane with droplets of octafluoropropane C_3F_8 (OFP) and vapor bubbles in the center without taking into account (1) and taking into account (2) phase transitions. As can be seen from Fig. 1, the presence of phase transitions leads to a decrease of the phase velocity (a) and an increase of the attenuation coefficient (b) at frequencies lower than the resonant frequency of the inclusions. Let us note that in the particular case for a mixture of water with vapor bubbles, a similar picture is observed [22–27]. Taking mass transfer processes into account leads to a decrease of the resonant frequency of bubbles and phase velocity, as well as an increase of the attenuation coefficient at pre-resonance frequencies.

Figure 2 shows the dependencies of the phase velocity (a) and the attenuation coefficient (b) for a mixture of polydimethylsiloxane with OFP droplets and vapor bubbles in the center at different relaxation times of the elastic phase. In recent work [28], it was shown that an increase in the relaxation time of the elastic carrier phase leads to less attenuation of the radial oscillations of the bubble and to its faster growth. Relaxation time is the time required for a viscoelastic medium or material to return to its initial unstressed state. Consequently, short relaxation times of the elastic liquid will suppress the radial movements of the bubble, since the carrier phase allows a rapid return to the initial unstressed state.

Figure 2 shows that taking into account the viscoelasticity of the carrier phase leads to an increase of the resonant frequency of inclusions. In this case, with a decrease in the parameter λ_l or when the carrier phase becomes more elastic, the resonant frequency of the bubbles increases, and the attenuation coefficient and the range of change of the phase velocity decrease. Calculations also showed that with increasing shear modulus G_l of the carrier medium, the resulting dispersion curves qualitatively coincide with the curves on Fig. 2, so they are not presented here. The above statements are in good agreement

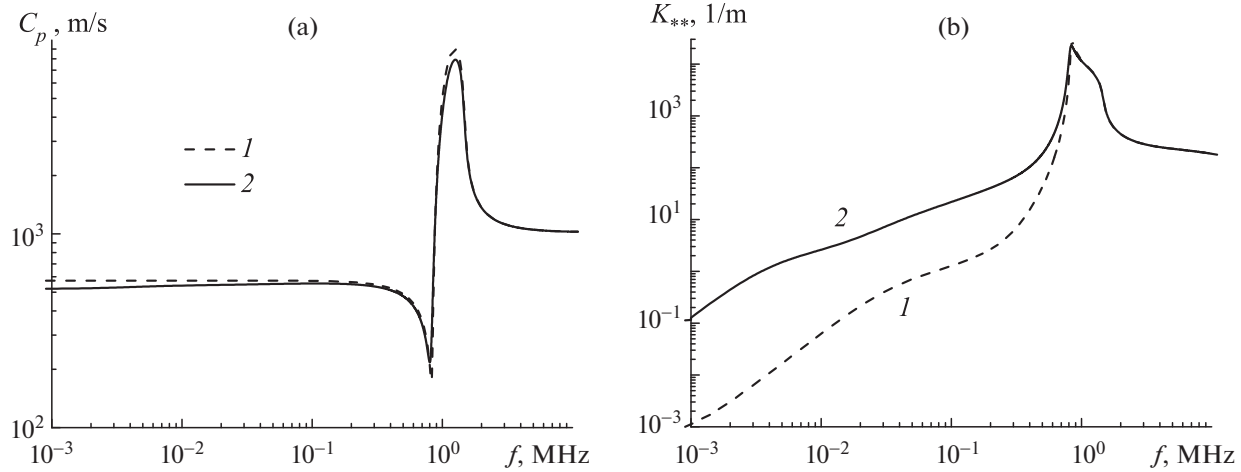


Fig. 1. Dependencies of the phase velocity (a) and attenuation coefficient (b) on the perturbation frequency for a mixture of polydimethylsiloxane with OFP droplets and vapor bubbles in the center: 1—without phase transitions, 2—taking phase transitions into account.

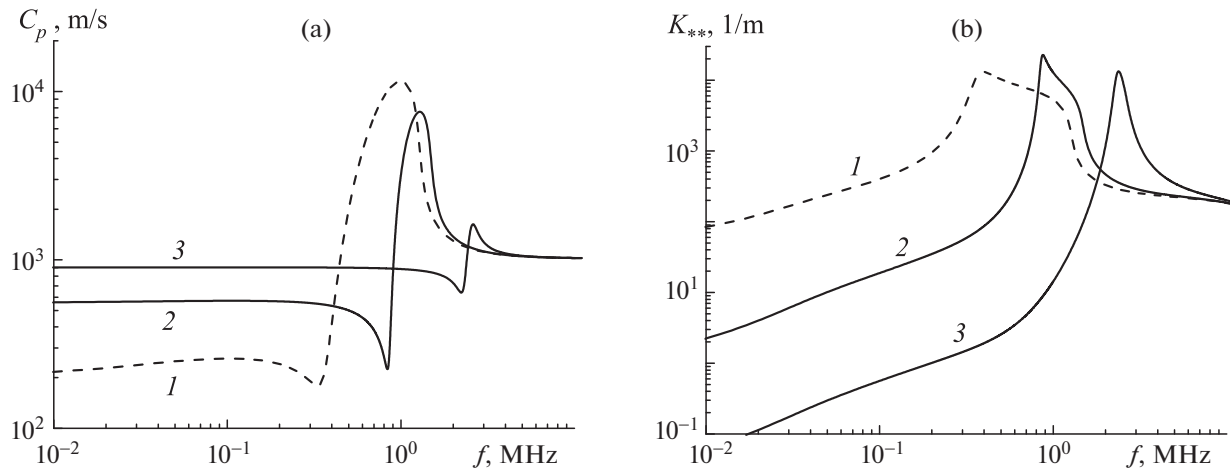


Fig. 2. Phase velocity (a) and attenuation coefficient (b) at different relaxation times of the elastic carrier phase: $\lambda_l =$ (1) 0, (2) 0.9, (3) 0.1.

with experiments [29, 30] for an elastic medium with gas bubbles and for water with gas bubbles covered with a rubber shell. A comparison of the theory with these experimental data was carried out in the work of the authors [31].

6. CONCLUSIONS

When an acoustic wave propagates in a viscoelastic liquid with perfluorocarbon droplets and vapor bubbles in the center, it has been established that phase transitions lead to a decrease of the phase velocity and an increase of the attenuation coefficient at frequencies lower than the resonant inclusion frequency. It has been established that an increase of the shear modulus of the elastic carrier medium leads to a shift of the dispersion curves towards an increase of the frequency of disturbances. It has been shown that as the relaxation time λ_l decreases or when the carrier phase becomes more elastic, the resonant frequency of the bubbles increases, the attenuation coefficient and the range of changing of the phase velocity decrease.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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